

**Physics as Kinematics of a Flowing Space:
Mass Anchoring, Vector Light Speed, and Electrodynamics from
a Single Space Field**

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Abstract

We present a postulate-based framework in which space itself is in motion, and all physics is the kinematics of a single space field $\mathbf{C}$ of fixed magnitude $|\mathbf{C}| = c$. The framework rests on four postulates: (P1) every spatial point carries a velocity vector $\mathbf{C}$ with $|\mathbf{C}|^2 = c^2$ (vector light speed — c is a property of space, not a speed limit for matter); (P2) photons are exactly comoving with the flow, so they carry no “anchoring” and are massless; (P3) mass is the anchoring effect of internal motion (spin) against the motion of space, so nonzero spin implies nonzero mass; and (P4) space moves in three directions, the normal direction emerging algebraically as $\sigma_3 = i\sigma_1\sigma_2$, which realizes the three-color/three-gluon structure of SU(3). Under these postulates the electromagnetic fields are kinematic derivatives of the space field, $E = -\partial_t C$ and $B = \partial_x C$; Faraday’s law becomes an identity, Ampere–Maxwell becomes equivalent to the wave equation of C , and the gauge redundancy of the vector potential is eliminated by the physical condition $|\mathbf{C}| = c$. Inhomogeneous flow yields the Gordon metric with $\Phi = \frac{1}{2}v^2$ in the weak-field limit, reproducing the Newtonian limit of general relativity; black holes appear as sinks of the flow whose horizon is the light-speed surface, so that inescapability is a corollary of the postulates rather than a dynamical result. The three-direction hypothesis predicts a glueball mass pattern $m = \sqrt{N} M_0$ that is consistent with lattice QCD (0^{++} at $\sqrt{3} M_0$ with $M_0 \approx 1$ GeV, 0.0% deviation), and the mode counts $N = 3, 6, 7$ align with the degeneracies $d(n) = \frac{1}{2}(n+1)(n+2)$ of the isotropic three-dimensional harmonic oscillator. All central theorems are formalized in the Lean 4 theorem prover (no axioms, no admitted proofs). Four further explorations are formalized and checked against data: spin as the emergence of the three-direction structure (SFS1–SFS5), double-slit interference as interference of space-helix harmonics (DS1–DS4), the fractal cosmos with the KBC void resolving the Hubble tension ($\delta^* = -0.249$ inside the observed range), and glueball three-direction coupling with a massification gradient; the glueball $\sqrt{N} M_0$ sequence with $N = 3, 6, 7$ falls inside the lattice ranges for all three channels. We state the epistemic status honestly: at the present stage the framework is a self-consistent *conceptual reconstruction* — mathematically and numerically indistinguishable from standard physics wherever it overlaps — and we catalogue the falsifiable predictions and open problems that would elevate it beyond reinterpretation.

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I. INTRODUCTION

Hilbert’s sixth problem asks for an axiomatic derivation of physics [1]. The present project — ProjectionPhysics, the physical extension of the Hidden-space Bridge System (HIBS) of Liu and Xu [2] — pursues a specific instance of this program. HIBS starts from a hidden space S of “hidden numbers” and a non-injective projection $\pi : S \rightarrow \mathbb{R} \oplus i\mathbb{R}$; all observables are functions of the image $\pi(S)$ and the kernel $\ker \pi$, and no third degree of freedom exists. The classical complex plane $\mathbb{C}$ is not fundamental; it is the observable slice of S under π .

ProjectionPhysics adds a physical layer to this algebraic skeleton, guided by a single intuition (Liu, 2026): *space itself moves*, and light speed is the equivalent speed of that motion. From this intuition four postulates are distilled (Sec. II), and from them three bodies of physics are derived:

1. **Mass.** Mass is not the excitation energy of a field; it is the *anchoring* of internal motion (spin) against the motion of space. Nonzero spin implies nonzero mass; exact comotion with the flow implies zero mass (Sec. IV).
2. **Electrodynamics.** The electromagnetic fields are *kinematic derivatives* of the space field C : $E = -\partial_t C$, $B = \nabla \times C$. Maxwell’s equations reduce to the wave equation of C ; Faraday’s law is an identity, and the gauge redundancy of the vector potential is killed by the physical condition $|\mathbf{C}| = c$ (Sec. V).
3. **Gravitation.** Inhomogeneous flow distorts the metric; the Gordon metric follows, the weak-field potential is $\Phi = \frac{1}{2}v^2$, and black holes are sinks of the flow whose horizon is the light-speed surface (Sec. VI).

Throughout, a methodological commitment is observed that is unusual for a theory paper: *every* central theorem is accompanied by a machine-checked proof in the Lean 4 theorem prover [3, 4], and every numerical claim by a reproducible script in the public repository [5]. This is not decoration: in a framework whose content is mostly at the level of algebraic identities, formal verification is what separates “claimed” from “established”.

We also commit, equally unusually, to an explicit *honesty contract* (Sec. IX B): a four-layer assessment of what each result actually is — mathematical identity, structural correspondence with known physics, numerical matching, or conceptual reinterpretation — and

we record, in writing, which real-discovery criteria are *not* satisfied. The reader will find the framework self-consistent, partially novel in its derivation route, and currently indistinguishable from standard physics at the level of formulas and numbers.

II. POSTULATES OF THE FLOWING-SPACE FRAMEWORK

A. Vector light speed (SLS1–SLS3)

Postulate II.1 (Vector light speed). *There exists a vector field $\mathbf{C}(x)$ on space with*

$$|\mathbf{C}(x)|^2 = c^2 \quad \forall x, \quad (1)$$

where c is the universal speed of light. c is therefore a property of space itself — the equivalent speed of space’s own motion — not a maximum speed for matter in space [6].

This is a genuine shift of ontology. In the standard view, c bounds the propagation of signals through space; here, c describes the motion of space, and matter moves *relative* to it. The relative velocity of matter with respect to the flow is

$$u = \frac{dx}{dt} - \mathbf{C}(x). \quad (2)$$

Postulate II.2 (Photons are comoving). *Photons are exactly comoving with the flow: $u = 0$. A photon’s worldline is a streamline of $\mathbf{C}$.*

Postulate II.3 (Mass is anchoring). *The mass of a particle is the anchoring effect of its internal motion against the motion of space: m is a norm of the spinor flow $\sigma\psi$ generated by the internal state,*

$$m := \|\sigma\psi\| \geq 0, \quad (3)$$

with $m = 0$ iff the internal state is trivial. Equivalently, in the hidden number representation, $m^2 = h^2$ for $h \in \ker \pi$.

Postulate II.4 (Three-direction motion). *Space moves in three directions: planar motion (two directions) plus motion along the normal vector, which is not an independent input but emerges algebraically from the planar generators,*

$$\sigma_3 = i \sigma_1 \sigma_2, \quad (\sigma_1 \sigma_2)^2 = -1, \quad \sigma_1 \sigma_2 + \sigma_2 \sigma_1 = 0. \quad (4)$$

The three directions (x, y, z) of space are identified with the three color directions of $SU(3)$: the normal vector $(1, 1, 1)$ is the hypercharge direction, and the color-singlet profile $(c_0, c_1, c_2) = (1, 1, 1)$ lies along it.

Postulate II.2 together with Postulate II.1 implies that a photon worldline satisfies $|dx/dt| = |\mathbf{C}| = c$, and hence the proper time

$$d\tau^2 = dt^2 - \frac{dx^2}{c^2} \quad (5)$$

vanishes identically along it: *photons do not experience time because they are the motion of space itself* (SM1). Massive particles deviate from the flow ($u \neq 0$), which is precisely their mass (SM3c): mass is *the inability to comove with space at the equivalent speed c* . Time itself becomes a measure of deviation from the flow: $d\tau^2 > 0$ iff $|dx| < c dt$ (SM2).

B. Relation to the HIBS axioms

The postulates are not independent of the HIBS algebra. Postulate 3 interprets the HIBS kernel $\ker \pi$ as the carrier of internal (hidden) motion; the Clifford generators σ_i are the representation of the projection algebra, and i itself emerges from the anticommutation of the planar generators, $(\sigma_1\sigma_2)^2 = -1$, rather than being an axiom. The Lean formalization makes this explicit: in `PauliMathlib.lean` and `SpaceLightSpeed.lean` the statements `normal_direction_emerges_from_plane` and `planar_motion_products_give_i` are proved from the anticommutation relations alone [6].

III. MATHEMATICAL STRUCTURE

A. The algebraic skeleton

The HIBS projection structure provides the vocabulary: a non-injective projection $\pi : S \rightarrow V$ with kernel $\ker \pi$, a quadratic form Q on the image, and a Clifford representation σ of the form. All theorems in this paper live on top of this skeleton; the Lean files `Definitions.lean`, `Kernel.lean`, `Completeness.lean`, and `Mass.lean` establish the elementary facts (projection pairs, null cones, information conservation) [5].

B. Spinor flow and anchoring

Let $\psi \in \mathbb{C}^2$ be a two-component spinor and σ_i the Pauli matrices. The *spinor flow* in direction i is $\sigma_i \psi$. The anchoring mass candidate is

$$m = \sum_i |(\sigma_i \psi)_0|^2 + |(\sigma_i \psi)_1|^2, \quad (6)$$

and the core theorems (MC1, Lean) state

$$\psi \neq 0 \implies m > 0, \quad \psi = 0 \implies m = 0. \quad (7)$$

The glueball minimal version (MC2) is a color-singlet triplet $G = (a, b, c)$ with profile $(1, 1, 1)$:

$$m_G^2 = |a|^2 + |b|^2 + |c|^2 > 0 \quad \text{whenever} \quad (a, b, c) \neq (0, 0, 0). \quad (8)$$

The number three is not accidental: the bridge theorem[7] (SLS5) equates the norm squared of the three-direction space motion (3) with the mass squared of the three-gluon singlet (3).

C. Spin-statistics and compositeness constraints

An exploration module `CliffordSix.lean` and `SpinStatistics.lean` verifies the well-known obstruction that pure glueballs cannot assemble into fermions (spin-statistics), and that the eight-dimensional Clifford spinor of $\text{Cl}(6)$ realizes the color octet, with the branching $\text{SU}(3) \supset \text{SU}(2)$, $3 \rightarrow 2 \oplus 1$ [5]. These modules are kept separate from the main line because their physical content is standard.

IV. MASS: ANCHORING OF INTERNAL MOTION

A. The electron and the photon

The framework assigns:

- **Photon:** no internal structure, exactly comoving ($u = 0$) $\implies$ zero anchoring $\implies m = 0$.
- **Electron:** spin means the normal-direction component $\sigma_3 \psi \neq 0$ (the normal motion trajectory of the spin) $\implies$ positive anchoring $\implies m > 0$.

The mechanism chain is fully formalized (MC5', SM3c, SM3d in Lean):

$$\psi \neq 0 \implies \sigma_3 \psi \neq 0 \implies m > 0 \implies dx \neq c dt \implies d\tau^2 > 0. \quad (9)$$

A structural picture of the electron as a local flow structure (“tornado”) is explored numerically: the electron is a swirl ($\text{curl } \nabla \times \mathbf{C} \neq 0$, providing the anchoring mass) combined with a sink/source (divergence $\nabla \cdot \mathbf{C} \neq 0$, providing the charge and the Coulomb $1/r^2$ field), while the photon is a propagating wave of the same flow. The classical self-energy of a point charge cut off at the flow scale $r_0 = 0.20 \text{ fm}$ evaluates to $U(r_0) = 3.60 \text{ MeV} \approx 7.05 m_e$, and at the classical electron radius $r_e = 2.82 \text{ fm}$ to $U(r_e) = 0.255 \text{ MeV} \approx 0.5 m_e$; the electron mass lies between the two estimates [8]. This is reported honestly as an order-of-magnitude coincidence, not a derivation: r_0 is a fitted input, and the classical 4/3 and 1/2 factor problems remain.

B. Glueball spectrum: $\sqrt{N} M_0$

The three-direction hypothesis makes a quantitative statement about the lightest glueball. With M_0 the mass scale of one spatial mode,

$$m_G(N) = \sqrt{N} M_0, \quad (10)$$

the lattice values of [9, 10] are matched as follows:

TABLE I. Glueball masses from the $\sqrt{N} M_0$ rule with $M_0 = 0.987 \text{ GeV}$ ($r_0 = \hbar c/M_0 = 0.200 \text{ fm}$), compared with lattice QCD [9, 10]. The fit parameter is M_0 ; N is the mode count.

Channel	N	$\sqrt{N} M_0 \text{ (GeV)}$	Lattice (GeV)	Deviation
0^{++}	3	1.710	1.71 ± 0.05	0.0%
2^{++}	6	2.418	2.40 ± 0.12	0.8%
0^{-+}	7	2.612	2.56 ± 0.15	2.0%

The primary match is the 0^{++} glueball: $\sqrt{3} M_0 = 1.710 \text{ GeV}$ against the lattice value $1.71 \pm 0.05 \text{ GeV}$, a 0.0% deviation with $M_0 \approx 1 \text{ GeV}$ — the natural nuclear scale — and $r_0 \approx 0.2 \text{ fm}$, the proton Compton wavelength scale.

C. Why $N = 3, 6, 7$: harmonic-oscillator degeneracies

The mode counts admit a first-principles candidate. The degeneracies of the isotropic three-dimensional harmonic oscillator are

$$d(n) = \frac{(n+1)(n+2)}{2} = 1, 3, 6, 10, 15, 21, \dots \quad (11)$$

and the lattice mode counts align:

$$0^{++} \rightarrow N = 3 = d(1), \quad 2^{++} \rightarrow N = 6 = d(2), \quad 0^{-+} \rightarrow N = 7 = d(3) - d(1). \quad (12)$$

Against the two-dimensional alternative $N = 2, 4, 5$, the three-direction interpretation wins by 2.75σ in the weighted deviation ($\sigma_A = 0.263$ vs. $\sigma_B = 3.012$). The physical picture is a glueball as a bound quantum of the three-direction space flow — a standing-wave combination of metric perturbations, with $m^2 \propto (\text{number of modes}) \times M_0^2$. Because the flow is geometric (volume-preserving, $\det g = -1/c^2$) rather than material, no Michelson–Morley-type anisotropy is induced.

Two honest caveats are recorded: (i) the rule $0^{-+} \rightarrow d(3) - d(1)$ is an ad-hoc symmetry subtraction (removing the modes with 0^{++} symmetry) and lacks a symmetry derivation; (ii) M_0 itself has no first-principles origin (it is the one fit parameter). The predictions that follow from the rule are listed in Sec. XA.

V. ELECTRODYNAMICS AS KINEMATICS OF THE SPACE FIELD

A. The central derivation

Let C denote the space field of Postulate II.1. Define the electromagnetic fields as *kinematic derivatives* of C [11]:

$$E = -\partial_t C, \quad B = \partial_x C \quad (1+1D), \quad B = \nabla \times C \quad (3D). \quad (13)$$

Then the following are immediate and formalized (MS1–MS5, Lean, zero admitted axioms):

1. **Faraday’s law is an identity** (MS2). With $E = -\partial_t C$ and $B = \partial_x C$,

$$\partial_t B = \partial_t \partial_x C = \partial_x \partial_t C = -\partial_x E, \quad (14)$$

where the middle step is the commutation of mixed partials (MS1). Faraday is not an independent law; it is the statement that C is a single field. In 3D the same argument gives $\partial_t B = -\nabla \times E$, and $\nabla \cdot B = \nabla \cdot (\nabla \times C) = 0$ is automatic.

2. **Ampere–Maxwell is the wave equation of C** (MS3).

$$\partial_t E = -c^2 \partial_x B \iff \partial_t^2 C = c^2 \partial_x^2 C. \quad (15)$$

The two sides are literally the same equation. Hence the independent content of the vacuum Maxwell system is a single wave equation for C ; E and B are its apparent forms.

3. **Sources are the driving term** (MS5). A charge/current J enters as a source:

$$\partial_t^2 C = c^2 \partial_x^2 C + J, \quad (16)$$

so charge is not an external “bullet” but the driving term of the space field equation itself. In the static limit, $C'' \approx -J$, verified numerically.

4. **Gauge redundancy is killed by physics** (MS4). In the standard vector-potential formulation, C would be the gauge-dependent potential A . Here C is physically fixed by Postulate II.1 ($|\mathbf{C}| = c$ everywhere): the gauge freedom is eliminated not by a choice of convention but by the physical condition that C is the actual velocity field of space.

In short: *electromagnetism is the kinematics of the space field*. Maxwell’s equations do not describe two fields with two independent laws; they describe one field with one wave equation, and the familiar electromagnetic laws are consequences of the definition of E and B as derivatives of C .

B. Consistency with the flow postulates

The identification $c = 1/\sqrt{\mu_0 \varepsilon_0}$ (MF2) is preserved: c is simultaneously the flow speed of space and the propagation speed of the C wave. Dispersion $\omega = \pm ck$ holds as a factorization $(\omega - ck)(\omega + ck) = 0$ (MF1), and the photon-comoving postulate (SLS2) implies $|v_{\text{photon}}| = |C| = c$, so $d\tau^2 = 0$ identically (MF4, self-consistent). A leapfrog FDTD simulation with

CFL= 1 propagates the C wave packet at exactly $1.000c$, and the Faraday and Ampere residuals are zero at machine precision for both analytic traveling waves and the numerical leapfrog solution [11].

C. What is genuinely new here

The kinematic content — that Maxwell’s equations are equivalent to the wave equation of the vector potential — is standard electromagnetic theory. What is repository-specific is the *physicalization* of the potential: C is not an auxiliary field but the velocity field of space, with the fixed magnitude $|\mathbf{C}| = c$ that eliminates gauge freedom. This is a conceptual step; its falsifiability is discussed in Sec. IX B.

VI. GRAVITATION AND STRONG-FIELD STRUCTURE

A. From momentum conservation to the Gordon metric

The route from the postulates to gravity follows the classical Gordon construction [12]: an inhomogeneous flow $v(x)$ distorts the metric. With $g = \text{diag}(1, -1/c^2)$ in the comoving frame, a flow along x gives the Gordon metric (SG1)

$$g_{\mu\nu} = \begin{pmatrix} 1 - v^2/c^2 & v/c^2 \\ v/c^2 & -1/c^2 \end{pmatrix}, \quad \det g = -\frac{1}{c^2}, \quad (17)$$

and the proper time (SG8)

$$d\tau^2 = dt^2 - \frac{(dx - v dt)^2}{c^2}. \quad (18)$$

The key quantitative result (SG11) is the weak-field match: with $g_{tt} = 1 - v^2/c^2$ from the flow and $g_{tt} = 1 - 2\Phi/c^2$ from standard weak field general relativity,

$$\Phi = \frac{1}{2}v^2, \quad (19)$$

so the Newtonian acceleration is

$$a = -\nabla \left(\frac{1}{2}v^2 \right) = -v \nabla v, \quad (20)$$

i.e., gravity is the inhomogeneity of the flow (SLS6). The full chain — inhomogeneous flow $\rightarrow$ Gordon metric $\rightarrow$ Christoffel symbols $\rightarrow$ geodesic equation $\rightarrow$ Newtonian limit — is

verified numerically with the generic formula $\Gamma_{\mu\nu}^\lambda = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$, including a correction pass that recomputed all eight Christoffel components [13]. Proper time remains positive for massive particles ($d\tau^2 > 0$) and vanishes for photons, whose worldlines satisfy $|dx - v dt| = c dt$ *relative to the flow* — the correct photon condition, as opposed to the comoving condition $dx = v dt$, which would incorrectly give $d\tau^2 = dt^2$ (a modeling pitfall recorded in the formalization) [13].

B. Black holes as sinks of the flow

The postulates (rather than a field equation) determine the strong-field picture. A black hole is a region where the flow converges to a point: $C = -c\hat{r}$ inside $r < r_h$ (a sink), with all streamlines terminating at the singularity [14]. The horizon is the light-speed surface (BH1):

$$g_{tt} = 0 \iff |v| = c. \quad (21)$$

Inescapability is then a *corollary of Postulate II.2*: photons are streamlines, and escaping would require moving against the flow, which contradicts exact comotion (PH2). No appeal to general relativity is needed. Numerically, $|C| = c$ holds to $< 10^{-12}$ along all streamlines, $d\tau^2 = 0$ to $< 10^{-12}$, and an outward photon freezes at the horizon ($dx/dt = 0$), the coordinate picture matching Schwarzschild. The white hole is the reversed flow (BH5–BH6: g_{tt} invariant, g_{tx} flips), and a wormhole is a sub-light flow tube whose throat has a peak speed $v_{\text{peak}} < c$ (WH1: at the throat $v = 0$ and the light cone is restored), through which a photon passes with $d\tau = 0$ throughout [15]. This is structurally identical to the Hamilton–Lisle river model [16].

VII. NUMERICAL AND COMPUTATIONAL VALIDATION

All numbers in this paper are produced by scripts in the repository [5]. Table II collects the verification results.

TABLE II. Computational validation summary. “Residual zero” means machine-precision zero.

Check	Result
C -wave packet speed (leapfrog FDTD, CFL= 1)	$1.000\,c$
Faraday residual (analytic / leapfrog)	0.0 / 0.0
Ampere–Maxwell residual (analytic / leapfrog)	0.0 / 0.0
Static charge drives C (MS5)	$C'' \approx -J$, residual 0.033
Glueball 0^{++} ($\sqrt{3}\,M_0$)	1.710 vs. 1.71 ± 0.05 GeV (0.0%)
Glueball 2^{++} ($\sqrt{6}\,M_0$)	2.418 vs. 2.40 ± 0.12 GeV (0.8%)
Glueball 0^{-+} ($\sqrt{7}\,M_0$)	2.612 vs. 2.56 ± 0.15 GeV (2.0%)
Horizon condition $g_{tt} = 0 \Leftrightarrow v = c$	exact
Outward photon at horizon	$dx/dt = 0$ (frozen)
Time dilation at horizon	$d\tau/dt \rightarrow 0$
Wormhole throat peak	$v = 0.8\,c < c$, no horizon
$d\tau^2$ along streamlines	$< 10^{-12}$
$ C' = c$ along streamlines	$< 10^{-12}$
CHSH: helix model S	2.0005 ± 0.004 (local bound 2)
CHSH: quantum prediction	$2\sqrt{2} = 2.8284$; Aspect 1982: 2.697
Electron self-energy $U(r_0 = 0.20\,\text{fm})$	$3.60\,\text{MeV} = 7.05\,m_e$
Electron self-energy $U(r_e = 2.82\,\text{fm})$	$0.255\,\text{MeV} = 0.50\,m_e$

A. Lean formalization

The theorem prover component is summarized in Table III. Every listed module compiles with zero `sorry` (no admitted axioms) and zero warnings, under Lean 4.

VIII. RECENT EXPLORATIONS: SPIN, DOUBLE-SLIT, FRACTAL COSMOS, AND GLUEBALL DYNAMICS

Four further checks of the framework were carried out after the first version of this paper; all are formalized in Lean (zero `sorry`) and validated numerically (artifacts and scripts in the repository).

TABLE III. Lean 4 formalization summary. Job counts are from `lake build` on the current revision.

Module (file)	Theorems
<code>SpaceLightSpeed.lean</code>	SLS1–SLS5: universal $ \mathbf{C} = c$; photon zero anchoring; electron positive anchoring; $\sigma_3 = i\sigma_1\sigma_2$ emergence; three-direction/three-gluon bridge
<code>MinimalCore.lean</code> (Archive)	MC1–MC2, MC1h: nonzero spin $\Rightarrow$ nonzero mass; $m_G^2 = a ^2 + b ^2 + c ^2$
<code>SpaceMetric.lean</code>	SM1–SM6: photon $d\tau = 0$; mass $\Leftrightarrow$ deviation from flow; metric consistency; $\det g = -1/c^2$
<code>SpaceGravity.lean</code>	SG1–SG11: Gordon metric; $\Phi = \frac{1}{2}v^2$; Newtonian limit
<code>MaxwellSpace.lean</code> (Explorations)	MS1–MS5: Faraday identity; Ampere $\Leftrightarrow$ wave equation; sources as driving terms (4131 jobs, zero sorry)
<code>MaxwellFlow.lean</code> (Explorations)	MF1–MF6, PH1–PH2: dispersion; $c = 1/\sqrt{\mu_0\varepsilon_0}$; inescapability as postulate corollary (4130 jobs)
<code>BlackHoleWormhole.lean</code> (Explorations)	BH1–BH7, WH1: horizon = light-speed surface; white hole; wormhole throat (4128 jobs)
<code>LorentzRebuild.lean</code>	LR1–LR5: boost preserves $\eta = \text{diag}(-1, 1)$; $\gamma^2 - \gamma^2\beta^2 = 1$; velocity addition; $\beta < 1$
<code>DiracMathlib.lean</code>	DB1'–DB6': $(\gamma^0)^2 = 1$, $(\gamma^i)^2 = -1$; mass = chiral coupling
<code>EntanglementHelix.lean</code> (Explorations)	EH1–EH4: CHSH local bound $ S \leq 2$ (Bell kernel)

A. Spin as the emergence of the three-direction structure

Within the three-direction postulate (P4), the spin structure is not an input: the three directions generate a Clifford algebra, and the imaginary unit emerges as the oriented volume element

$$i = \sigma_1\sigma_2\sigma_3, \quad [\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k, \quad S^2 = \frac{3}{4}\hbar^2 = s(s+1)\hbar^2 \quad (s = \frac{1}{2}). \quad (22)$$

The value $s = \frac{1}{2}$ is the Casimir value forced by the 2-dimensional (irreducible) representation of $C\ell(3)$, and the sign flip under a 2π rotation (“fermionic” behaviour) is the double-cover geometry of $SU(2)$ over $SO(3)$ — not a quantum postulate. This answers, within the framework, the question Dirac left open (where does the “complex intrinsic property” come from? — it is the orientation of the three directions) while honestly recording that the choice of the 2-dimensional representation and the value of $\hbar$ remain inputs (“second input” gap).

B. Double-slit interference as interference of space-helix harmonics

Photons are comoving with the flow (P2); if the flow carries a helical structure (circular motion in the plane plus translation along the normal, the two planar modes of P4), then the projection of the helix onto any plane is a harmonic wave ($x = R \cos \omega t$; the xy projection is exactly a circle, $x^2 + y^2 = R^2$, DS1). Interference fringes are then the algebraic consequence of adding two harmonics,

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}, \quad I(\theta) = 4I_0 \cos^2\left(\frac{\pi d \sin \theta}{\lambda}\right), \quad (23)$$

and observation (which-slit information) removes the interference cross term $2 \cos \Delta\varphi$ in $|z_1 + z_2|^2$ — the “compression of the space harmonics” — leaving two bands (DS4). The mathematics is identical to standard wave optics and decoherence; the re-interpretation is conceptual (the wave is the projection of the space helix, the particle is the compressed-harmonic localisation).

C. Fractal cosmos, the KBC void, and the Hubble tension

If cosmic structure is fractal (observed galaxy distribution $D \simeq 1.2-1.5$), then in the flow picture the expansion is the divergence of the flow, $H(\mathbf{x}) = H_0(1 - \delta(\mathbf{x})/3)$ (top-hat linear theory), and voids are the necessary empty regions of the fractal “skeleton” (cytoskeleton analogy). The local Hubble constant measured inside a deep void is elevated: reproducing the observed tension $\Delta H/H = (73.0 - 67.4)/67.4 = 8.3\%$ requires a void depth

$$\delta^* = -3 \Delta H/H = -0.249, \quad (24)$$

which falls inside the observed KBC void range $\delta \in [-0.3, -0.2]$; a simulated fractal flow field (96^3 FFT, Poisson-solved, spectral divergence identity exact to machine precision) produces

a largest void with central $\delta = -0.27$ and an 9% Hubble boost, consistent with the 5.6 km/s/Mpc tension. This reproduces the known KBC-void resolution of the tension (Keenan et al. 2013, Shanks et al. 2019) in the flow language; it is a consistency check, not a new prediction.

D. Glueball three-direction coupling and the massification gradient

Describing the glueball in the three-direction version of the modified Maxwell system ($V = g C_1 C_2 C_3$ three-linear coupling) yields a stable localised bound state whose mass grows monotonically with the coupling; the mass-squared decomposes into the three-direction gradient sum $m^2 = \sum_i (\partial_x C_i)^2$ (equal gradients $\Rightarrow m \propto \sqrt{3} g$, the same $\sqrt{3}$ factor as the mode rule), and the “gradient that resists the space flow” is the massification mechanism ($\Phi = \frac{1}{2} v^2$, SG11). With the mode rule of Sec. IV ($N = 3, 6, 7$), the updated lattice comparison (Morningstar–Peardon 1999; Chen et al. 2006) gives

TABLE IV. Glueball masses from the $\sqrt{N} M_0$ rule with $M_0 = 0.93$ GeV (central), updated with the three-direction coupling check.

Channel	N	$\sqrt{N} M_0$ (GeV)	Lattice (GeV)	Verdict
0^{++}	3	1.61	1.475–1.75	inside
2^{++}	6	2.28	2.15–2.40	inside
0^{-+}	7	2.46	2.30–2.60	inside

All three channels fall inside the lattice ranges (previously 2^{++} was 3–13% low with $N = 5$; the oscillator degeneracy $N = 6 = d(2)$ brings it inside). The inverse-derived $M_0 \simeq 0.93 \pm 0.07$ GeV is consistent across the three channels to better than 7%. The N -sequence and the M_0 calibration remain fits (second-input gap), and the coupling model is a 1+1-dimensional toy, so the verdict is: data consistent at the order-of-magnitude level; first-principles prediction not yet achieved.

IX. RELATION TO EXISTING PHYSICS

A. What is standard, restated

Where the framework touches existing physics, the formulas and numbers agree with it: the Gordon metric is a classical construction [12]; the river model of black holes is Hamilton and Lisle [16]; Maxwell’s equations as the wave equation of a potential is textbook; $d\tau^2 = 0$ for photons is Minkowski; the glueball spectrum is lattice QCD [9, 10]. The framework currently contains *no* experimental number that standard physics does not also produce.

B. Four-layer assessment

For each result we record its status on four layers [17]:

1. **Mathematical identity.** True by construction: e.g., the Faraday identity, Ampere $\Leftrightarrow$ wave equation, $g_{tt} = 0 \Leftrightarrow |v| = c$. These are rigorous (machine-checked) but content-free as physics.
2. **Structural correspondence.** Known physics restated: the vector-potential formulation, the Gordon/river geometry, the spin-statistics obstruction.
3. **Numerical matching.** Consistent but not new: the glueball $\sqrt{N}M_0$ rule has one fit parameter M_0 ; the electron self-energy coincidence sits between $0.5 m_e$ and $7 m_e$ with a fitted cutoff.
4. **Conceptual reinterpretation.** Unfalsifiable at present: “gravity is inhomogeneous flow”, “mass is anchoring”, “ c is the speed of space” — the explanation layer changes while the formulas do not.

The one result that rises above layer 1 is the Maxwell-space derivation (Sec. V): although the algebraic kernel is the standard vector potential formulation, the *physicalization* of the potential ($|\mathbf{C}| = c$) and the reduction of Maxwell’s system to a single wave equation of a physically fixed field is repository-specific content. Even here, the honest statement is that it is a derivation-route novelty, not a new prediction.

X. DISCUSSION

A. Predictions

The framework makes the following falsifiable statements.

1. **P1 — inside the horizon, light inevitably falls into the singularity.** This follows from the comoving postulate and is verified numerically [14].
2. **P2 — gravitational redshift**

$$\frac{\omega_2}{\omega_1} = \frac{c - v_2}{c - v_1} \quad (25)$$

with the flow profile $v(r)$ as the single free input. The direction agrees with GR; numerical agreement with GR requires the flow profile to take the GR shape, which the repository does not derive from first principles.
3. **P3 — transverse electromagnetic modes vanish inside the horizon** (the field becomes purely radial) [14].
4. **P4 — isotropy is preserved.** Since $|\mathbf{C}| = c$ in every direction, light speed is isotropic by construction; this is structurally untestable, and we say so.
5. **Glueball spectrum.** The harmonic-oscillator mode rule predicts, for $n = 4, 5, 6$,

$$m = \sqrt{15}, \sqrt{21}, \sqrt{28} \times M_0 = 3.82, 4.52, 5.22 \text{ GeV}, \quad (26)$$

and, if the $d(n) - d(1)$ subtraction rule holds, $n = 4$ gives $\sqrt{12} M_0 = 3.42 \text{ GeV}$. Lattice data on $1^{++}, 2^{-+}, 3^{++}$ channels can discriminate.

6. **Entanglement under inhomogeneous flow.** In the entanglement-helix exploration [18], a gravitational (inhomogeneous-flow) phase shift would modify the correlation function $E(\Delta)$; the local bound $|S| \leq 2$ is proven, and the helix geometry saturates it ($S = 2.0005 \pm 0.004$), while the quantum value $2\sqrt{2}$ requires Born-rule readout — an honest place where the framework must either produce Born’s rule or fail.

B. Open problems

The following gaps are structural, not cosmetic:

- **The $966\times$ gap.** The electron anchoring mechanism has the right direction (nonzero spin $\Rightarrow$ nonzero mass) but the magnitude, at the flow scale r_0 , overshoots m_e by a factor of order 10^3 ; every candidate formula collapses to an identity ($m \equiv M_0/2$). An independent input not of order $\hbar/r_0$ is needed.
- **Origin of M_0 .** The glueball fit parameter has no first-principles derivation; connecting it to a confinement scale requires QCD dynamics the repository does not have.
- **The N -sequence.** $(3, 6, 7)$ is not yet derived; the $d(3) - d(1)$ subtraction for 0^{-+} is ad hoc.
- **Continuum 3D calculus.** The 3D discrete curl/divergence identities are now formalized (SF1–SF5: $\text{divcurl} = 0$ automatic, 3D Faraday automatic, gradient curl-free); the continuum version and continuity remain open.
- **The N -sequence.** $(3, 6, 7)$ now has a dynamical candidate (three-linear coupling, GC1–GC4) but is still not derived; the $d(3) - d(1)$ subtraction for 0^{-+} remains ad hoc.
- **Representation choice and $\hbar$.** Why the electron uses the 2-dimensional representation, and the numerical value of $\hbar$, have no origin in the framework (second-input gap).
- **Missing standard content.** Charge quantization, the value of e , QCD confinement, the complete curvature tensor and the Einstein field equations, and Born’s rule are all outside the framework.
- **Lorentz violation?** The only physical exit from pure reinterpretation is inhomogeneous flow inducing Lorentz violation; no experimental scheme or magnitude estimate exists yet.

XI. CONCLUSION

We have presented a postulate-based framework — space in motion, $|\mathbf{C}| = c$ — from which mass, electrodynamics, and gravitation emerge as kinematics. The framework is internally consistent, every central claim is machine-checked in Lean 4, and every number

is reproducible from the repository. Its distinctive content is the physicalization of the vector potential ($|\mathbf{C}| = c$) and the consequent reduction of Maxwell’s equations to the wave equation of a single space field, plus the three-direction realization of $SU(3)$ with its $\sqrt{N}M_0$ glueball pattern.

Its epistemic status is stated without inflation: at present this is a *conceptual reconstruction* of known physics — rigorous, coherent, and indistinguishable from standard physics in formulas and numbers. Real discovery criteria (first-principles derivation of M_0 , resolution of the $966\times$ gap, a new falsifiable prediction with an experimental scheme) are not yet met. The predictions of Sec. X A are the tests that would change this verdict.

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Appendix A: Lean formalization overview

The formalization lives in the public repository `github.com/logos-42/Hibs-Physics` (directory `ProjectionPhysics/`). Conventions: theorems are named by module prefix (MS, MF, BH, SG, SM, SLS, MC, LR, EH); a prime denotes a mathlib rewrite of an earlier core proof; (Explorations) modules are newer-direction results kept separate from the main line. Every file compiles under `lake build` with zero `sorry` and zero warnings; the job counts in Table III are from the current revision.

Appendix B: Proof sketches

1. Faraday’s law is an identity (MS2)

In 1+1 dimensions, with $E = -\partial_t C$ and $B = \partial_x C$:

$$\partial_t B = \partial_t \partial_x C \stackrel{(\text{MS1})}{=} \partial_x \partial_t C = -\partial_x E.$$

In 3D, $B = \nabla \times C$ gives $\partial_t B = \nabla \times \partial_t C = -\nabla \times E$, and $\nabla \cdot B = \nabla \cdot (\nabla \times C) = 0$ identically. Both directions are formalized; the commutation step (MS1) is the only ingredient.

2. Ampere–Maxwell is the wave equation (MS3)

Using the definitions,

$$\partial_t E + c^2 \partial_x B = -\partial_t^2 C + c^2 \partial_x^2 C,$$

so $\partial_t E = -c^2 \partial_x B$ holds iff $\partial_t^2 C = c^2 \partial_x^2 C$. The equivalence is a direct substitution (Lean: MS3, both directions).

3. Horizon = light-speed surface (BH1)

For the Gordon metric $g_{tt} = 1 - v^2/c^2$,

$$g_{tt} = 0 \iff v^2 = c^2 \iff |v| = c.$$

Inside the horizon $|v| > c$ gives $g_{tt} < 0$ (BH7, role swap seed).

4. Weak-field potential (SG11)

Equating the flow expression and the GR expression for g_{tt} ,

$$1 - \frac{v^2}{c^2} = 1 - \frac{2\Phi}{c^2} \implies \Phi = \frac{v^2}{2},$$

and the geodesic equation yields $a = -\nabla\Phi = -v \nabla v$ (Newtonian limit).

5. CHSH local bound (EH1–EH2)

For dichotomic observables $A_a, B_b \in \{\pm 1\}$,

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b'),$$

and $|S| \leq 2$ follows from $A_a B_b + A_a B_{b'} + A_{a'} B_b - A_{a'} B_{b'} = A_a(B_b + B_{b'}) + A_{a'}(B_b - B_{b'}) \in \{\pm 2\}$, since each parenthesis is 0 or ± 2 . This is the algebraic kernel of Bell’s theorem [19]; the helix geometry saturates the bound numerically ($S = 2.0005 \pm 0.004$).

6. Glueball mass positivity (MC2)

For $G = (a, b, c)$ with $m_G^2 = |a|^2 + |b|^2 + |c|^2$, if $(a, b, c) \neq (0, 0, 0)$ then at least one $|a_i|^2 > 0$, hence $m_G^2 > 0$; conversely $m_G^2 = 0$ forces $a = b = c = 0$. Trivial in content, but it is the formal anchor of “three directions give mass”.